Venkata Lakshmi Parimala Pasupuleti

Senior Python Developer · Enterprise Applications · Data-Intensive Systems · Oracle · PostgreSQL · AWS · CI/CD · Automated Testing · ML Ecosystem · Distributed Backend · Automation

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PROFESSIONAL SUMMARY

Senior Python Developer with 8 years designing, building, and operating enterprise-grade applications and data-intensive systems in mission-critical production environments. At Entrata, owned end-to-end delivery of 20+ production modules at 99.5% uptime across 100K+ monthly active users - Oracle, PostgreSQL, and MySQL databases with 200+ query optimizations (30% performance gain), secure coding with 100% PHI compliance, and automation pipelines eliminating 80% of manual operational workload. Built AgentProbe - a Python-native automated testing and observability platform with CI-integrated regression gating across 1,240+ test cases - and applied NumPy, Pandas, scikit-learn, and PyTorch in ML research pipelines at George Mason University (GPA 3.87, M.S. Data Analytics Engineering, May 2026).

TECHNICAL SKILLS

Python & Backend: Python (8yr enterprise), SQL, PL/SQL, FastAPI, PHP, JavaScript, TypeScript, Bash, Docker, Microservices, Scalable Distributed Systems, AWS (S3, EC2, RDS, Glue, DynamoDB), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions), Automated Testing, Secure Coding, Oracle, PostgreSQL, MySQL, MS SQL, Linux, Git

Data & ML Ecosystem: NumPy, Pandas, SciPy, scikit-learn, PyTorch, PySpark, Feature Engineering, Statistical Modeling, Data Pipelines, ETL, Schema Design, Query Tuning, Execution Plan Analysis, Indexing, Tableau, R

AI & Automation: LangGraph, LangChain, RAG Pipelines, LLM Orchestration, ChromaDB, Model Evaluation, Observability, HITL, Prompt Engineering, OpenAI API, Anthropic API, Agile/Scrum

EXPERIENCE

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built enterprise-grade ML pipelines (Python, R) applying NumPy, Pandas, scikit-learn, and statistical modeling on WHO and UK Road Safety datasets - $R^2 = 0.977$; full pipeline from ingestion through feature engineering, model optimization, and validation across 190+ countries.
- Mentoring 600+ students in Python, SQL, and database design; designed dashboards improving research comprehension by 40%.

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built Python automation platform (LangGraph) for mission planning - end-to-end pipeline from data ingestion through automated validation and conditional routing; automated testing enforcing compliance constraints before delivery.
- Implemented RAG pipeline with observability on retrieval quality; performance-optimized for low-latency across distributed data sources.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Designed and operated enterprise-grade Python applications end-to-end** - 20+ production modules at 99.5% uptime across 120+ Agile sprints; zero compliance violations across 7 years; 100K+ monthly active users in mission-critical affordable housing SaaS.
- **Database engineering and query optimization** - Oracle, PostgreSQL, MySQL production databases; 200+ SQL and PL/SQL query optimizations (30% performance gain) through execution plan analysis, indexing strategy, partitioning, and stored procedure refactoring; schema design across multi-tenant production environments.
- **Automation and secure coding at scale** - Python/Bash automation pipelines eliminating 80% of manual operational workload; 100% PHI compliance with Libsodium-encrypted data pipelines (zero security incidents); scalable ETL processing fragmented federal data into 180,000 domain records per release cycle.
- **Distributed systems and CI/CD** - Docker microservices redesign (50% scalability improvement, 30% deployment reduction); RabbitMQ event-driven processing; Redis caching; GitHub Actions CI/CD with automated testing and release pipelines; AWS S3, EC2, RDS, Glue at production scale.
- Mentoring 5+ engineers; 300+ peer code reviews; debugging and root-cause analysis across complex enterprise production systems.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built enterprise Python and PHP backend modules (PostgreSQL, Oracle); data-intensive pipeline for 1.6M HUD/Tax Credit records with validation and reconciliation; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Python Automated Testing & Observability Platform Python · Testing

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Enterprise-grade Python automation platform: 15-endpoint FastAPI service, 5-metric automated testing engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate blocking quality failures before merge, LLM-as-judge scoring, and multi-provider routing. Automated test harness across 1,240+ test cases - zero regressions shipped to production.

Leasel: Distributed Python Agentic Backend Python · Distributed

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Scalable Python distributed backend: 8-agent stateful application with conditional routing, multi-source data ingestion, and reliability controls. 93.6% precision, 87.9% F1 across 1,240 automated test cases; CI quality gate with telemetry-driven scoring.

AI on FHIR: NLP-Driven Medical Records Query System Python · NLP

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · NumPy · Pandas | github.com/venkatalakshmiparimala/AI_on_FHIR

Python NLP platform translating natural-language queries into FHIR API calls using BERT embeddings and LLMs - reduced retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017